

Monotone bounded invertibility is strictly weaker than the monotone limit property

Autonomous mathematical discovery packet
Run fa_banach_001, lane 0, agent_lane_00

12 August 2026

Status. Candidate full counterexample, likely valid, pending human review. The proof is exact and has no computational dependency.

1 Source question

Mironchenko, Kawan, and Glück [1] study, on an ordered Banach space (X, X^+) , the discrete-time inequality

$$x(k+1) \leq A(x(k)) + u(k). \quad (1)$$

They say that (1) has the *monotone limit property (MLIM)* if some $\xi \in \mathcal{K}_\infty$ has the following property: for every $\varepsilon > 0$, every constant input $u(k) \equiv w \in X^+$, and every decreasing solution, there is an N such that

$$\|x(N)\| \leq \varepsilon + \xi(\|w\|).$$

They say that $\text{id} - A$ has the *monotone bounded invertibility (MBI)* property if some $\xi \in \mathcal{K}_\infty$ satisfies

$$(\text{id} - A)(v) \leq w \implies \|v\| \leq \xi(\|w\|) \quad (v, w \in X^+). \quad (2)$$

After proving $\text{MLIM} \Rightarrow \text{MBI}$, they ask whether MBI is strictly weaker than MLIM or whether the two properties are equivalent. Figure 1 records the exact passage.

Proposition 5.3 *Let (X, X^+) be an ordered Banach space and let $A : X^+ \rightarrow X^+$ be a nonlinear operator. If system (5.1) has the MLIM property, then the operator $\text{id} - A$ has the MBI property.*

Proof Assume that $(\text{id} - A)(v) \leq w$ for some $v, w \in X^+$. We write this as $v \leq A(v) + w$. Hence, $x(\cdot) \equiv v$ is a constant solution of (5.1) corresponding to the constant input sequence $u(\cdot) \equiv w$. By the MLIM property, there exists $\xi \in \mathcal{K}_\infty$ (independent of v, w) so that for every $\varepsilon > 0$ there is N with

$$\|v\|_X = \|x(N)\|_X \leq \varepsilon + \xi(\|u\|_\infty) = \varepsilon + \xi(\|w\|_X).$$

Since this holds for every $\varepsilon > 0$, we obtain $\|v\|_X \leq \xi(\|w\|_X)$, which completes the proof. $\square$

Whether the MBI property is strictly weaker than the MLIM property, or whether they are equivalent, is an open problem. In the next proposition, though, we show that they are equivalent under certain assumptions on the operator A or the cone X^+ . Later, in Propositions 7.16 and 7.17 we show their equivalence for linear operators and for the gain operator $\Gamma_\otimes$ with linear gains, defined on $\ell_\infty(I)$.

The cone X^+ is said to have the *Levi property* if every decreasing sequence in X^+ is norm-convergent [1, Def. 2.44(2)]. Typical examples are the standard cones in L^p -spaces for $p \in [1, \infty)$, and the standard cone in the space c_0 of real-valued sequences that converge to 0. We note in passing that if the cone X^+ has the Levi property, then it is normal [1, Thm. 2.45].

Figure 1: The open question in arXiv:2007.05705, page 12.

2 Proof intuition

The missing phenomenon is escape without norm convergence. In ℓ_∞^+ , successive tail indicators form a decreasing norm-one sequence. A linear shift propagates these tails, but fails MBI because shallow ramps are approximate fixed points. We replace its scalar edge map by a bistable nonlinear gain: values near zero contract quadratically, the value one propagates exactly, and large values contract affinely. Small forcing is then trapped near zero, which gives MBI, while an unforced tail of ones still travels to infinity and violates MLIM.

3 Counterexample

Let $X = \ell_\infty(\mathbb{N})$ with its usual norm and positive cone. Define

$$g(t) := \begin{cases} t^2, & 0 \leq t \leq 1, \\ (t+1)/2, & t \geq 1, \end{cases} \quad (Ax)_1 := 0, \quad (Ax)_{n+1} := g(x_n). \quad (3)$$

Theorem 1. *The map g belongs to $\mathcal{K}_\infty$, and $A : \ell_\infty^+ \rightarrow \ell_\infty^+$ is a globally 2-Lipschitz monotone operator. Moreover:*

- (i) $\text{id} - A$ has the MBI property;
- (ii) the system (1) does not have the MLIM property.

Consequently, MBI is strictly weaker than MLIM, even for globally Lipschitz monotone max-type gain operators on ℓ_∞^+ whose nonzero edge gains belong to $\mathcal{K}_\infty$.

Proof. The two formulas in (3) agree at one. Each is strictly increasing, and their union is strictly increasing; also $g(0) = 0$ and $g(t) \rightarrow \infty$. Thus $g \in \mathcal{K}_\infty$. Since the slopes are bounded

in absolute value by two, g is 2-Lipschitz. It follows coordinatewise that A is well-defined, monotone, and 2-Lipschitz on ℓ_∞^+ .

We prove MBI directly. Suppose $v, w \in \ell_\infty^+$ and $(\text{id} - A)(v) \leq w$, and put $\delta = \|w\|_\infty$. Coordinatewise,

$$v_1 \leq \delta, \quad v_{n+1} \leq g(v_n) + \delta \quad (n \geq 1). \quad (4)$$

First assume $0 \leq \delta \leq 1/4$ and set

$$\alpha_\delta := \frac{1 - \sqrt{1 - 4\delta}}{2} = \frac{2\delta}{1 + \sqrt{1 - 4\delta}}.$$

Then $\delta \leq \alpha_\delta \leq 1/2$, $\alpha_\delta^2 + \delta = \alpha_\delta$, and $\alpha_\delta \leq 2\delta$. Induction in (4) gives $v_n \leq \alpha_\delta$ for every n : the base case follows from $v_1 \leq \delta$, and if $v_n \leq \alpha_\delta \leq 1$, then

$$v_{n+1} \leq v_n^2 + \delta \leq \alpha_\delta^2 + \delta = \alpha_\delta.$$

Hence $\|v\|_\infty \leq 2\delta$.

Now assume $\delta \geq 1/4$ and put $B_\delta = 1 + 2\delta$. Since $B_\delta \geq 1$,

$$g(B_\delta) + \delta = \frac{B_\delta + 1}{2} + \delta = B_\delta.$$

As $v_1 \leq \delta \leq B_\delta$, monotonicity of g and induction in (4) yield $v_n \leq B_\delta$ for all n . Therefore $\|v\|_\infty \leq 1 + 2\delta$.

Define

$$\xi(r) := \begin{cases} 6r, & 0 \leq r \leq 1/4, \\ 1 + 2r, & r \geq 1/4. \end{cases} \quad (5)$$

The formulas agree at $r = 1/4$, where both equal $3/2$. Thus ξ is continuous, strictly increasing, unbounded, and $\xi(0) = 0$, so $\xi \in \mathcal{K}_\infty$. The preceding two bounds imply $\|v\|_\infty \leq \xi(\delta)$, proving (2).

It remains to disprove MLIM. For $k \geq 0$, let

$$x(k) = (\underbrace{0, \dots, 0}_{k \text{ entries}}, 1, 1, 1, \dots).$$

Since $g(0) = 0$ and $g(1) = 1$, we have $x(k+1) = A(x(k))$. Hence x is a zero-input solution of (1). It is decreasing in the cone order, but $\|x(k)\|_\infty = 1$ for every k . If MLIM held with any $\xi \in \mathcal{K}_\infty$, then applying it with $w = 0$ and $\varepsilon = 1/2$ would give some N with

$$1 = \|x(N)\|_\infty \leq \frac{1}{2} + \tilde{\xi}(0) = \frac{1}{2},$$

a contradiction.

Finally, A has the max-type gain-operator form: the only nonzero scalar gain entering coordinate $n+1$ is $\gamma_{n+1,n} = g$. This proves the strengthened scope assertion. $\square$

4 Why the positive results in the source do not apply

The standard positive cone of ℓ_∞ is normal and generating, but it does not have the Levi property: the decreasing sequence $x(k)$ above is not norm convergent. The operator A is not compact, because the sequence $A(x(k)) = x(k+1)$ has no norm-convergent subsequence. Thus the example lies exactly beyond the two additional hypotheses in Proposition 5.4 of [1]. It is nonlinear, so it also does not conflict with the source's equivalence theorem for positive linear operators.

5 Verification and novelty status

The companion script checks the scalar fixed-point identities, both invariant intervals, and long finite prefixes of the tail trajectory. The proof itself is exact and does not depend on those computations.

A bounded search through 12 August 2026 covered the exact arXiv id and title; the phrases “MBI property”, “MLIM property”, “monotone bounded invertibility”, and “monotone limit property”; and related later small-gain papers. The current source of Kawan’s arXiv:2503.03925v6 [2] was inspected because it studies closely related infinite-network MBI conditions. No source located states this counterexample or resolves the exact additive MBI–MLIM question. Novelty confidence is moderate pending expert review.

6 Human-review recommendation

High priority. Verify the scalar invariant-interval argument, continuity of the piecewise function (5), and the exact quantifiers in Definitions 5.1–5.2 of [1]. If confirmed, the example gives a full and unusually simple negative answer while showing that the Levi/compactness mechanism in Proposition 5.4 is essential.

References

- [1] A. Mironchenko, C. Kawan, and J. Glück, *Nonlinear small-gain theorems for input-to-state stability of infinite interconnections*, Mathematics of Control, Signals, and Systems 33 (2021), 573–615; arXiv:2007.05705.
- [2] C. Kawan, *The Small-Gain Condition for Infinite Networks Modeled on ℓ^∞ -Spaces*, arXiv:2503.03925v6 (2026).